What is the Influence of Task Order in Continual Learning? An Empirical Study and Order-aware Evaluation for Motion Forecasting

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***Abstract*—Sensitivity to task order complicates the evaluation of continual learning: an algorithmic advantage observed along one learning history may not persist under another. We formulate algorithm comparison as an expectation over an explicit distribution of task orders and develop an evaluation framework based on shared sequences, uniform reference scores, and order-resolved performance. A descriptor of successive shifts in task distributions defines a supplementary weighting policy that is common to all compared algorithms. Experiments on motion forecasting reveal ranking reversals across orders and substantial order-dependent variation relative to differences between algorithm averages. Analysis of the starting task and the reference term in backward transfer further distinguishes relative improvement from absolute prediction quality. The descriptor associations are evaluated empirically within the selected task set. These results support evaluation protocols that quantify sensitivity to task order and state the sequence distribution underlying aggregate comparisons.**

***Index Terms*—Continual learning, order-aware evaluation, motion forecasting.**

I. Introduction

CONTINUAL learning (CL) addresses sequential knowledge acquisition under changing data distributions while maintaining performance on previously encountered tasks [1]–[3]. Parameter updates for a new task can interfere with representations required by earlier tasks. Replay, regularisation, and gradient constraints mitigate this interference through different retention mechanisms [4]–[8]. Their performance consequently depends on the learning history as well as the task set.

Several evaluation protocols repeat prescribed task sequences while varying initialisation or within-task sample order [8], [9]. Such repetitions characterise stochastic training variation conditional on the sequence. They do not quantify performance variation across task permutations, which studies of order-sensitive forgetting and order fairness examine separately [10], [11]. Algorithms evaluated on the same task set may change their relative ranking when the presentation order changes. A comparison restricted to one sequence therefore provides conditional evidence of an algorithmic advantage.

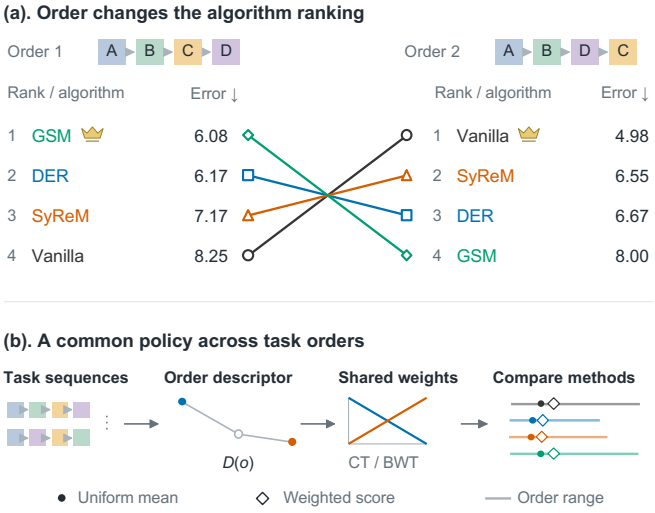


Fig. 1. Task order and algorithm comparison. (a) Mean current-task FDE (m) and rankings for two sequences. (b) A common order policy relates uniform means, weighted scores, and ranges across orders.

For N distinct tasks, the order space contains $N!$ permutations [12]. Each permutation defines a sequence of distribution transitions and a corresponding learning history. Studies of order-dependent forgetting and task similarity demonstrate that these transitions influence retention and transfer [10], [13], [14]. Evaluation consequently requires a specified order set and an aggregation policy, in addition to a fixed learning protocol. This requirement is consistent with prior recommendations to assess continual forecasting under varying data-stream sequences [15]. Measures of order-wise performance disparity address sensitivity across sequences [11]; a distribution over orders further specifies the relative contribution of each sequence to an aggregate algorithm comparison.

Existing studies analyse effects of task order, derive ordering principles, or estimate sequence transferability [10], [12], [14]. Our focus is the evaluation of algorithms under a common order distribution. Uniform averages define the equal-

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order reference, whereas order-resolved outcomes identify ranking reversals and performance dispersion. We supplement this reference with a descriptor-based policy derived from changes in successive distances between task distributions. Applying identical weights to every algorithm permits comparison under an explicit emphasis on particular transition patterns. Fig. 2 connects task distributions, sequential training, performance measurement, and aggregation under this shared policy.

Motion forecasting provides the empirical setting, with sequential adaptation across traffic environments and a shared prediction backbone [16], [17]. The experiments evaluate the complete order space of the selected task set with repeated training for each algorithm–sequence combination. Fig. 1 shows a ranking reversal between two orders that differ in their final two tasks. Analysis of the starting task and a decomposition of backward transfer examine the role of initial learning quality in the resulting comparisons.

The main contributions are summarised as follows:

1) We formulate algorithm evaluation over a distribution of task permutations, distinguishing stochastic training variability from variation across learning orders. The formulation defines a common comparison target for all evaluated algorithms.
2) We develop a descriptor of transition direction and a shared weighting policy that complement uniform-order evaluation. The resulting scores quantify the effect of an explicitly specified order emphasis on algorithm comparisons.
3) We characterise order-dependent ranking reversals among sequential fine-tuning and three memory-based learning strategies in motion forecasting. Adjustment for the starting task and decomposition of backward transfer identify how initial-reference effects contribute to the observed associations.

II. Related Work

A. Task Order and Curriculum Effects

Curriculum learning investigates the effect of example or task presentation on optimisation [18]–[20]. In CL, ordering also affects retention of previous knowledge. Bell and Lawrence [10] examine order-dependent forgetting and task-distance heuristics; Li and Hiratani [14] derive ordering principles in a teacher–student model. Analyses of task similarity, overparameterisation, and learned representations further characterise interference [13], [21].

Yoon *et al.* [11] quantify order-wise performance disparity, establishing order fairness as an evaluation criterion. Measures of sequence transferability relate task transitions to performance using learned predictors [12]. Our analysis instead uses marginal task statistics to define a shared comparison policy. Its purpose is to evaluate algorithm rankings across learning histories, complementing methods that select a curriculum or predict sequence transferability.

B. Continual Learning for Motion Forecasting

Continual forecasting addresses adaptation across changing traffic environments [15]. Approaches include generative replay [22]–[24], regularisation under distribution shift [25], and incremental prediction models [26]. Task-free continual forecasting considers streams without supplied task identifiers or boundary annotations [9], [27], [28].

D-GSM uses traffic divergence and scenario-organised memory to constrain gradient updates [16]. H2C combines diversity-based selection with equiprobable replay inspired by hippocampal circuits [9]. SyReM integrates a memory-loss constraint with gradient-based rehearsal selection [17], while Dual-LS coordinates long- and short-term knowledge through complementary rehearsal mechanisms [29]. These designs address adaptation and retention. The present study examines sequence-dependent evaluation using selected memory-based strategies and a shared forecasting backbone.

C. Evaluation Protocols for Continual Learning

Average performance, current-task performance, and backward transfer describe distinct aspects of sequential learning [6], [30], [31]. Their interpretation depends on the task stream, memory budget, and training protocol. Hyperparameter selection on the evaluation scenario can bias comparisons [32], and unequal update costs can alter conclusions about online CL [33].

Order coverage constitutes a further protocol choice. We distinguish repeated training within a sequence from evaluation across sequences, and assess backward transfer alongside initial and final error. This separates relative improvement from absolute prediction quality. Descriptor-weighted scores supplement these outcomes with an explicit order policy without modifying the underlying performance measures.

III. Problem Formulation

A. Sequential Learning Conditioned on Tasks

Let $[N] = \{1, \dots, N\}$ index a task set $\mathcal{T}$. Task j is specified by a distribution $\mathcal{D}_j$ on $\mathcal{X} \times \mathcal{Y}$ and a finite training set $\mathcal{S}_j = \{(\mathbf{X}_{jk}, \mathbf{Y}_{jk})\}_{k=1}^{n_j}$. The tasks share the input and output spaces, while their data distributions may differ. An order $o = (o_1, \dots, o_N)$ belongs to the permutation group $\mathfrak{S}_N$, with $|\mathfrak{S}_N| = N!$. Evaluation uses a common subset $\mathcal{O} \subseteq \mathfrak{S}_N$ for every algorithm.

For algorithm $a \in \mathcal{A}$, let $z_r^{(o,a,s)}$ denote its state after stage r in training replicate s. This state comprises model parameters, optimiser state, and algorithm-specific persistent information, including replay memory. Sequential learning is represented by

$$z_r^{(o,a,s)} = \mathcal{U}_a\big(z_{r-1}^{(o,a,s)}, \mathcal{S}_{o_r}; \xi_r^{(o,a,s)}\big), \tag{1}$$

where $\mathcal{U}_a$ is the update operator and $\xi_r^{(o,a,s)}$ denotes the stochastic inputs for that run and stage. The initial state is likewise run-specific. The memory component satisfies $|\mathcal{M}_r^{(o,a,s)}| \le B_a$ for a specified record capacity B_a; a record may include a stored model output. Historical training examples are accessible only through this memory. Task boundaries are provided, and all tasks use a shared prediction head. The evaluation does not impose a single-pass training requirement.

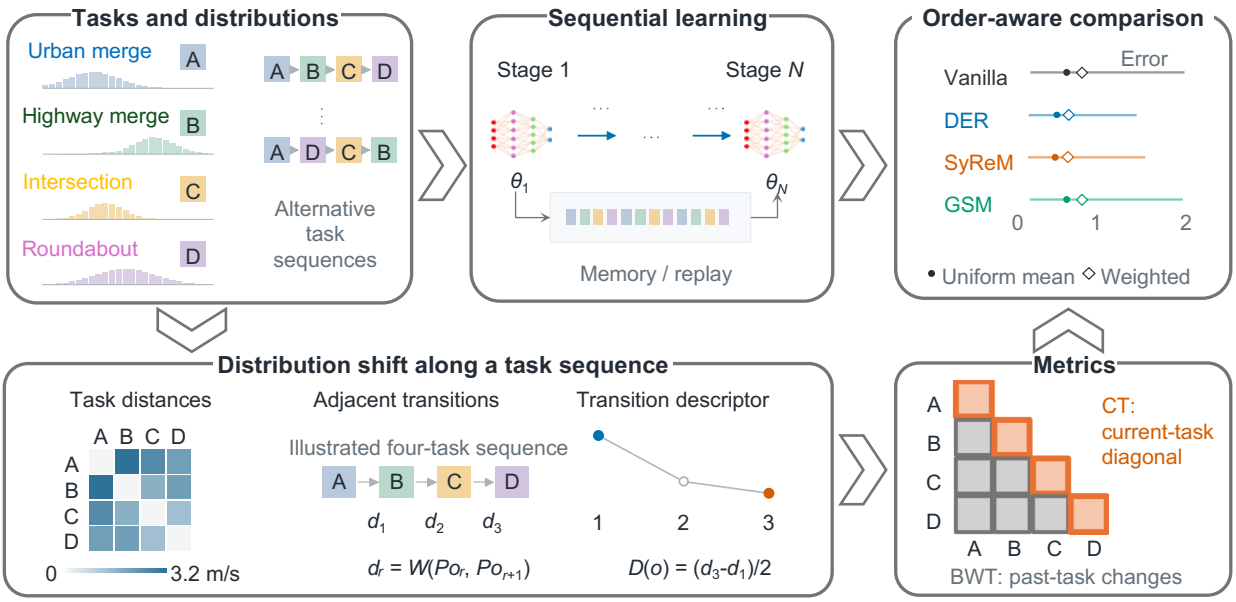


Fig. 2. Order-aware evaluation workflow. Task distributions define transition descriptors; sequential training provides current-task and backward-transfer outcomes. A shared order policy aggregates these outcomes for algorithm comparison. The distributions, networks, and comparison marks are schematic.

B. Risk and Knowledge Retention

For loss ℓ and predictor $f_{\boldsymbol{\theta}}$, the risk on task j is

$$\mathcal{R}_j(\boldsymbol{\theta}) = \mathbb{E}_{(\mathbf{X},\mathbf{Y})\sim\mathcal{D}_j}[\ell(f_{\boldsymbol{\theta}}(\mathbf{X}), \mathbf{Y})]. \tag{2}$$

A reference objective for stage $r > 1$ is

$$\boldsymbol{\theta}_r^\star \in \arg\min_{\boldsymbol{\theta}} \left\{ \mathcal{R}_{o_r}(\boldsymbol{\theta}) + \frac{\lambda_r}{r-1} \sum_{i<r} \mathcal{R}_{o_i}(\boldsymbol{\theta}) \right\}, \tag{3}$$

where $\lambda_r \geq 0$ balances current-task fit and past-task risk. Stage 1 contains only the current-task term. This objective specifies the desired retention–adaptation balance; it is not an assertion that every method optimises the same loss. Replay, regularisation, and gradient constraints approximate historical risk through different mechanisms [4]–[6], [8]. Absolute historical risk and its change since initial learning are distinct quantities, motivating separate measures of final error and backward transfer below.

C. Evaluation over Task Orders

Let $Z_m(o, a, \xi)$ be the scalar outcome of a complete run under metric m, where ξ collects initialisation and subsequent training randomness for the specified order and algorithm. For an order policy q satisfying $q(o) \geq 0$ and $\sum_{o\in\mathcal{O}} q(o) = 1$, the comparison target is

$$\mu_m(a; q) = \sum_{o\in\mathcal{O}} q(o)\, \mathbb{E}_\xi[Z_m(o, a, \xi)]. \tag{4}$$

For S training replicates, define $y_m(o, a) = S^{-1} \sum_{s=1}^{S} Z_m(o, a, \xi^{(o,a,s)})$. The empirical policy score is

$$\widehat{\mu}_m(a; q) = \sum_{o\in\mathcal{O}} q(o) y_m(o, a). \tag{5}$$

All algorithms share $\mathcal{O}$, q, and the metric definition. For fixed a, m, and q, assume Z_m has a finite second moment and draw $o \sim q$. The law of total variance separates the two sources of variability:

$$\begin{aligned}\mathrm{Var}(Z_m) = {} & \mathbb{E}_{o\sim q}[\mathrm{Var}_\xi(Z_m \mid o)] \\ & + \mathrm{Var}_{o\sim q}(\mathbb{E}_\xi[Z_m \mid o]).\end{aligned} \tag{6}$$

The first term describes stochastic training variation within orders; the second describes variation in expected performance across orders. Repetition of a single sequence addresses only the first component. Order-aware evaluation therefore requires both an explicit order policy and outcomes resolved by sequence. This definition does not assume independence among time steps within a trajectory or matched random initialisations across orders.

IV. Experimental Design

A. Motion Forecasting Tasks and Backbone

Each forecasting task comprises trajectory records from one category of road layout in INTERACTION [34]. We group twelve location subsets into four scenario-defined tasks:

- *A, Urban Merge*: vehicles negotiate lane convergence and zipper merging at the German MT site.
- *B, Highway Merge*: ramp-entry and lane-convergence interactions at the Chinese ZS highway site, represented by the archived ZS0 and ZS2 subsets.
- *C, Intersection*: crossing and turning interactions at four US intersection subsets, MA, GL, EP0, and EP1.
- *D, Roundabout*: entry, circulation, and exit interactions at EP, SR, and FT in the US, OF in Germany, and LN in China.

The site codes identify recording locations; ZS0 and ZS2 retain the local archive's subdivision. The corresponding task

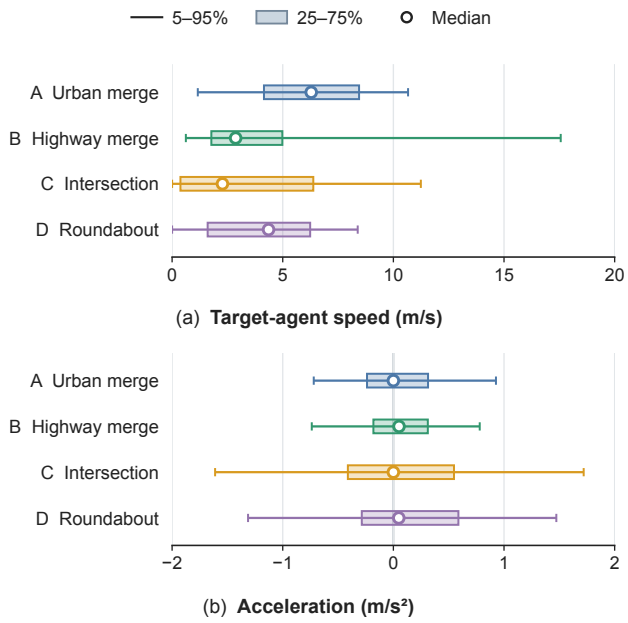


Fig. 3. Task feature distributions. Dots, bands, and whiskers indicate medians, interquartile ranges, and 5th–95th percentiles. Acceleration is restricted to $[-5, 5]$ m/s^2.

arrays contain 5,378, 50,905, 134,269, and 103,540 trajectory records. These counts describe stored records rather than independent time-step observations.

Fig. 3 compares the locations and spreads of the empirical target-agent feature distributions. Urban merging has the highest median speed (6.29 m/s), whereas highway merging combines a lower median (2.87 m/s) with the largest 95th percentile (17.56 m/s). Intersections have the lowest median speed (2.27 m/s). Intersections and roundabouts exhibit wider acceleration interquartile ranges than either merging task. The shared feature scales distinguish typical motion from distributional spread across the four scenarios.

The forecasting input is $\mathbf{X} = (\mathbf{X}_{\mathrm{TA}}, \mathbf{X}_{\mathrm{SA}}, \mathbf{M})$, containing the target-agent history, surrounding-agent histories, and map context. The ground-truth output $\mathbf{Y} \in \mathbb{R}^2$ is the target endpoint. UQnet [35] combines agent and map encoders with a spatial prediction head to produce a heatmap over future endpoints. The predictor samples W candidate endpoints $\{\hat{\mathbf{Y}}_w\}_{w=1}^{W}$ from that heatmap. The training loss follows the heatmap-based forecasting formulation in the backbone work.

Fig. 4 details the encoder and prediction branches. Lane attention encodes the map into $\mathbf{H}_{\mathrm{M}}$, an MLP embeds spatial coordinates as $\mathbf{H}_{\mathrm{C}}$, and a trajectory subnet extracts target-vehicle and surrounding-agent features, $\mathbf{H}_{\mathrm{TV}}$ and $\mathbf{H}_{\mathrm{S}}$. The interaction module fuses vehicle and map features, which the decoder combines with coordinate features to predict an endpoint heatmap. The auxiliary branch applies two cross-attention blocks to map, target-vehicle, and coordinate features to provide regularisation during training; it is removed at inference [35]. All compared algorithms use the same predictor. Trajectory completion [36] can extend endpoint predictions to full trajectories, but is not evaluated here.

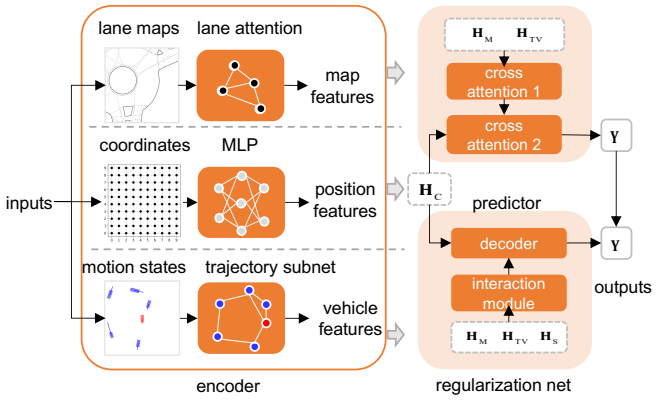


Fig. 4. UQnet backbone: feature encoding, interaction modelling, endpoint prediction, and auxiliary regularisation during training.

TABLE I
PUBLISHED MECHANISMS OF THE COMPARED ALGORITHMS.

Method	Stored examples	Output matching	Scenario partition	Gradient constraint	Gradient retrieval
Vanilla	×	×	×	×	×
DER	✓	✓	×	×	×
GSM	✓	×	✓	✓	×
SyReM	✓	×	×	✓	✓

B. Algorithms and Experimental Coverage

We compare four learning strategies under a common prediction architecture (Table I):

- *Vanilla*: sequential optimisation on the current task without replay, providing a reference for unconstrained adaptation.
- *DER* [8]: rehearsal of stored inputs with output matching to preserve previous predictions. The saved experiments use a forecasting adaptation of the original classification method.
- *GSM* [16]: the equal-allocation variant of D-GSM, with scenario-organised memory and gradient constraints derived from previous scenarios.
- *SyReM* [17]: gradient-based rehearsal selection combined with an average-memory loss constraint, coupling selective replay with gradient projection.

The selection spans output distillation, scenario-wise constraints, and selective constrained rehearsal, with Vanilla providing the no-memory reference. All memory-based runs use the recorded capacity of 2,000. Table I contrasts the cited designs. Stored examples support output matching, rehearsal, or the estimation of gradient constraints; scenario partition denotes allocation by past scenario. Training configurations are maintained separately from the released analysis.

The experiment evaluates all $4! = 24$ orders of $N = 4$ tasks, with four algorithms and three training replicates per combination (288 runs). Testing every task after each stage yields 1,152 stage logs and 4,608 stage–task records. Replicate labels do not imply matched initial states across orders. Evaluation data

and analysis code are available online[1]; resources for model training are maintained separately.

C. Forecasting and Continual Learning Metrics

Minimum final displacement error (FDE) measures the distance from the ground-truth endpoint to the closest predicted candidate:

$$\mathrm{FDE} = \min_{1\le w\le W} \|\hat{\mathbf{Y}}_w - \mathbf{Y}\|_2. \tag{7}$$

Miss rate (MR) reports the percentage of evaluated cases classified as misses by the experiment evaluator. We use its saved FDE and MR values; the present analysis does not recompute prediction-level metrics. The endpoint tolerance is part of that evaluator, rather than a parameter of the order-aware analysis.

For either error metric, let $E_{r,j}^{(o,a,s)}$ denote the result on task identity j after stage r. We reorder columns by presentation position and define

$$R_{r,i}^{(o,a,s)} = E_{r,o_i}^{(o,a,s)}. \tag{8}$$

Thus $R_{r,r}$ always evaluates the task just learned, even when its identity is not r. For clarity, we suppress the superscripts when defining the summaries:

$$\mathrm{CT} = \frac{1}{N}\sum_{r=1}^{N} R_{r,r}, \tag{9}$$

$$\mathrm{BWT} = \frac{1}{N-1}\sum_{r=2}^{N}\frac{1}{r-1}\sum_{i=1}^{r-1}(R_{r,i} - R_{i,i}), \tag{10}$$

$$\mathrm{Final} = \frac{1}{N}\sum_{i=1}^{N} R_{N,i}. \tag{11}$$

CT measures error upon initial learning. BWT averages subsequent error changes over stages: positive values indicate degradation and negative values improvement relative to the initial fit, reversing the accuracy-based sign convention. FDE is in metres, MR in percent, and MR-BWT in percentage points. Lower CT and Final indicate better absolute predictions; lower BWT indicates greater relative improvement and need not imply lower final error.

D. Descriptors of Task Distributions

We describe each task through empirical distributions of target-agent speed and acceleration. Speed is the norm of the two velocity components at each of nine historical time steps. Acceleration is the adjacent speed difference divided by 0.1 s, retaining values in $[-5, 5]$ m/s^2. States are pooled within each scenario group to form the empirical distributions. Correlated time-step observations contribute to these distributions, while training repetitions remain the unit for reporting stochastic variation.

The candidate measures probe complementary aspects of distribution shift. Wasserstein-1 quantifies displacement of probability mass along the physical feature axis [37]. KL divergence measures directed probability mismatch, whereas Jensen–Shannon (JS) divergence provides a symmetric, bounded comparison of the same histograms [38]. Maximum mean discrepancy (MMD) compares kernel mean embeddings without histogram discretisation [39]. Applying all four measures to speed holds the feature representation fixed while varying the notion of discrepancy. Applying Wasserstein-1 additionally to acceleration examines a different aspect of motion while retaining the same distance functional. These five combinations provide a structured sensitivity analysis, rather than an exhaustive search for an optimal task metric.

For feature $f \in \{\text{speed}, \text{acceleration}\}$, let F_i^f denote its empirical cumulative distribution function in task i. We compute

$$d_{ij}^{\mathrm{W}}(f) = \int_{\mathbb{R}} |F_i^f(t) - F_j^f(t)|\,\mathrm{d}t. \tag{12}$$

This calculation uses every retained observation, without subsampling or standardisation. Its units are m/s for speed and m/s^2 for acceleration.

For KL and JS, let c_{ib} count speed observations in bin b of $H = 100$ shared equal-width bins spanning the pooled range. With $\epsilon = 10^{-10}$, the smoothed probabilities are

$$p_{ib} = \frac{c_{ib}/\sum_{h=1}^{H} c_{ih} + \epsilon}{1 + H\epsilon}. \tag{13}$$

Writing $m_{ijb} = (p_{ib} + p_{jb})/2$, the divergences are

$$d_{ij}^{\mathrm{KL}} = \sum_{b=1}^{H} p_{ib}\log\frac{p_{ib}}{p_{jb}}, \tag{14}$$

$$d_{ij}^{\mathrm{JS}} = \frac{1}{2}\sum_{b=1}^{H}\left(p_{ib}\log\frac{p_{ib}}{m_{ijb}} + p_{jb}\log\frac{p_{jb}}{m_{ijb}}\right). \tag{15}$$

Natural logarithms are used. KL retains its direction; JS denotes the divergence, not its square root.

For MMD, a fixed sample of $n = 2{,}000$ speed observations per task is standardised using the pooled mean μ_v and standard deviation σ_v:

$$z_{iu} = \frac{v_{iu} - \mu_v}{\sigma_v}, \qquad k(z, z') = \exp[-(z - z')^2]. \tag{16}$$

The same samples are reused across pairs. The empirical squared discrepancy includes diagonal terms,

$$Q_{ij} = \frac{1}{n^2}\sum_{u,v=1}^{n} k(z_{iu}, z_{iv}) + \frac{1}{n^2}\sum_{u,v=1}^{n} k(z_{ju}, z_{jv}) - \frac{2}{n^2}\sum_{u,v=1}^{n} k(z_{iu}, z_{jv}), \tag{17}$$

$$d_{ij}^{\mathrm{MMD}} = \sqrt{\max(0, Q_{ij})}. \tag{18}$$

The clipping removes numerical roundoff below zero. This is a descriptive discrepancy; no kernel two-sample test is performed. Wasserstein-1, JS, and MMD matrices are symmetric with zero diagonals. Sampling details are given in Appendix A; scaling to $[0, 1]$ is applied only to plotted matrices.

The primary descriptor uses Wasserstein-1 distances between speed distributions, providing an interpretable scale for changes in motion. This choice emerged from an exploratory

[1]https://github.com/lzrbit/task_order

scan of 20 distance representations and 15 aggregation operators against the CL outcomes. We report the five candidate combinations under fully specified estimation settings and retain the original choice without further outcome-based retuning. Associations on this benchmark therefore describe the selected representation; independent task sets are required to validate its predictive use.

V. Order-aware Evaluation

A. A Descriptor of Transition Direction

For order o, let $\Delta_r(o) = d_{o_r,o_{r+1}}$ be the distance at the rth transition. The descriptor is its least-squares slope against transition position:

$$D(o) = \frac{\sum_{r=1}^{N-1}(r-\bar{r})\Delta_r(o)}{\sum_{r=1}^{N-1}(r-\bar{r})^2}, \qquad \bar{r} = N/2. \tag{19}$$

It is defined for $N \geq 3$. For the four-task experiment this reduces exactly to

$$D(o) = \frac{d_{o_3,o_4} - d_{o_1,o_2}}{2}. \tag{20}$$

The middle transition has zero coefficient. A positive value means the final transition is larger than the first; it does not imply that all transitions increase, or that individual tasks become intrinsically harder. We therefore refer to D as a descriptor of transition direction. Its interpretation as objective-specific difficulty is an empirical association to be checked, not part of the definition.

For symmetric distances, $D(o^{\mathrm{rev}}) = -D(o)$, where o^{rev} reverses the order. There are only three partitions of four tasks into two unordered pairs. Each partition gives two signed contrasts, so Eq. (20) has at most six distinct values across the complete order set. Swapping the tasks within either endpoint pair leaves D unchanged even though it can change training. These ties are retained in the correlations and weighting. Order-resolved results reveal the variation among sequences that share a descriptor value.

The slope is equivariant to a common positive affine transformation of all off-diagonal distances: if $d'_{ij} = ad_{ij} + b$ for $a > 0$, then $D'(o) = aD(o)$. Thus its rank and the min–max weights below are unchanged by a common linear rescaling. This statement does not hold for an arbitrary monotone transformation of the pairwise distances, because the slope is a difference of distances.

B. Order Policies and Algorithm Rankings

For algorithms a and b, define the measured difference $\delta_m(o; a, b) = y_m(o, a) - y_m(o, b)$. Equation (5) gives

$$\widehat{\mu}_m(a; q) - \widehat{\mu}_m(b; q) = \sum_{o\in\mathcal{O}} q(o)\delta_m(o; a, b). \tag{21}$$

A common sign of δ_m across orders preserves the weak ranking under every policy supported on $\mathcal{O}$. Opposite signs permit a ranking reversal under policies concentrating on different sequences. Order-wise differences thus distinguish uniform dominance from dependence on the evaluation distribution. This property concerns the measured means; replicate variability remains relevant to close comparisons. The uniform and descriptor-based policies below use the same runs and metric definitions.

C. Uniform and Descriptor-weighted Summaries

The uniform policy assigns equal probability to each evaluated sequence, giving

$$\bar{y}_m(a) = \frac{1}{|\mathcal{O}|}\sum_{o\in\mathcal{O}} y_m(o, a). \tag{22}$$

This score estimates performance under an equal-order evaluation policy. It is reported alongside the minimum and maximum order means, per-order ranks, and training variation. A deployment-specific comparison requires a correspondingly specified order distribution.

For a supplementary weighting policy, let $D_{\min} = \min_{o\in\mathcal{O}} D(o)$ and $D_{\max} = \max_{o\in\mathcal{O}} D(o)$, and define

$$\widetilde{D}(o) = \frac{D(o) - D_{\min}}{D_{\max} - D_{\min}}. \tag{23}$$

When $D_{\max} = D_{\min}$, uniform weights are used and the weighted result equals Eq. (22). Otherwise, the observed directions of association motivate the common weights

$$w_{\mathrm{CT}}(o) = 1 - \widetilde{D}(o), \qquad w_{\mathrm{BWT}}(o) = \widetilde{D}(o), \tag{24}$$

and, for each metric separately,

$$\bar{y}^{w}_m(a) = \frac{\sum_o w_m(o) y_m(o, a)}{\sum_o w_m(o)}. \tag{25}$$

The normalised weights define $q_m(o) = w_m(o)/\sum_{o'} w_m(o')$ in Eq. (5). The same policy applies to every algorithm. CT and BWT use opposite directions because their empirical associations with D have opposite signs. We do not combine FDE, MR, CT, and BWT into a single score with incompatible units.

The policy emphasises descriptor regions associated with larger errors or less favourable backward transfer. It does not provide a worst-case guarantee: a large realised error can receive a small or zero weight. Uniform means and order-resolved ranges are therefore retained alongside weighted scores.

D. Selection, Comparisons, and Scope of Inference

Descriptor selection and weighting direction were informed by this task set. Spearman correlations with the 16 algorithm–metric series use average ranks for ties and provide descriptive, rather than independent confirmatory, evidence.

Starting-task dependence is assessed by excluding A-first orders and by correlating ranked variables after removing starting-task group means. Residualisation removes additive group differences; remaining position-dependent interactions preclude a causal interpretation.

We also compare paths with the largest and smallest total transition distance,

$$L(o) = \sum_{r=1}^{N-1} d_{o_r,o_{r+1}}, \tag{26}$$

within each fixed starting task. Path length and slope are different summaries: L uses all three transitions, whereas D omits the middle one. These path comparisons probe order dependence conditional on the starting task and are not independent validation of the slope descriptor.

VI. Experimental Results

A. Order-resolved Algorithm Comparisons

Tables II and III report all orders. Each entry is a mean and sample standard deviation across three replicates. Order IDs follow lexicographic order of A–D, starting with A–B–C–D and ending with D–C–B–A. The full sequences are supplied with the machine-readable data.

Task order accounts for 71.3%, 64.6%, 76.3%, and 91.3% of the total centred sum of squares in FDE-CT, MR-CT, FDE-BWT, and MR-BWT, respectively. Algorithm effects account for 1.3%, 1.7%, 0.9%, and 3.4%, and order–algorithm interactions for 11.3%, 10.3%, 9.7%, and 1.3%. The remaining 16.0%, 23.4%, 13.1%, and 4.0% arise from training variation within order–algorithm combinations. This descriptive decomposition uses all 288 runs per metric and treats replicate numbers as local labels rather than a matched factor across combinations (Appendix A).

For FDE-CT, the algorithm-averaged order means range from 0.97 to 6.92 m, a spread of 5.95 m. The median gap between the best and worst algorithm within an order is 0.87 m, while the gap between the largest and smallest uniform-order algorithm means is 0.62 m. These comparisons show that changing the order can have a larger numerical effect than changing the algorithm, although part of that order effect reflects the starting task.

All four algorithms have the smallest mean FDE-CT on at least three orders: DER and SyReM each lead on nine, and Vanilla and GSM on three. All four also lead on at least one MR-CT order. For the BWT metrics, three algorithms attain a per-order minimum; DER does not. Thus the winner depends both on the order and on the metric. These counts use unrounded replicate means and do not establish that the winner is statistically distinguishable from the next method.

In Fig. 1, GSM has the smallest mean FDE-CT on A–B–C–D, whereas Vanilla leads on A–B–D–C. The orders share their first two task identities. Replicate standard deviations in Table II quantify the training variability associated with these comparisons.

B. Descriptor Associations and Starting-task Effects

Fig. 5 compares the five candidate task-distance matrices. Distances between the same task pair vary substantially with the feature and functional, which is one reason the descriptor choice must be reported. Table IV gives their associations with CL performance, with tied ranks preserved. MMD and speed Wasserstein induce the same ordering of descriptor values and therefore identical Spearman correlations. Their distance magnitudes and resulting weights differ. The correlation comparison thus supports both representations equally at the rank level; the primary analysis retains the interpretable speed-Wasserstein scale.

Across the 16 algorithm–metric series, absolute Spearman correlations range from 0.53 to 0.81. The algorithm-averaged correlations are -0.64 for FDE-CT, -0.63 for MR-CT, $+0.61$ for FDE-BWT, and $+0.74$ for MR-BWT. Fig. 6 resolves these associations for each algorithm and metric, showing performance across the six tied descriptor levels. Variation among observations at a shared level is part of the result, not numerical error: these orders share the endpoint contrast while differing elsewhere in the sequence.

Table V compares marginal and conditional associations. Excluding the six A-first orders reduces the mean absolute correlation in every metric. Removing all four starting-task group means from the ranked variables leaves associations of -0.48, -0.49, $+0.29$, and $+0.62$, respectively. The reduction is particularly large for FDE-BWT. This analysis shows that the identity of the starting task accounts for part of the observed association; it does not separate task size, scene type, initial optimisation quality, or their interactions.

A sensitivity comparison with symmetrised historical Wasserstein estimates produces the same descriptor ranking and hence the same Spearman correlations. The full-distribution calculation changes the distance magnitudes, however, so rank agreement alone does not imply identical weighting. Both distance versions are retained in the analysis package.

C. Backward Transfer and the Initial Reference

The large negative BWT values for A-first orders motivate a decomposition of the initial reference. Expanding Eq. (10) gives

$$\mathrm{BWT} = P - B, \tag{27}$$

where

$$P = \frac{1}{N-1}\sum_{r=2}^{N}\frac{1}{r-1}\sum_{i<r} R_{r,i}, \tag{28}$$

$$B = \frac{1}{N-1}\sum_{r=2}^{N}\frac{1}{r-1}\sum_{i<r} R_{i,i}. \tag{29}$$

Both P and B are error levels. The first is the average subsequent error on previously seen tasks; the second is the corresponding weighted initial reference. A high B can make BWT very negative even when P is not especially low. The first task is also referenced at every later stage. For $N = 4$, its coefficient in B is $11/18$, compared with $5/18$ and $1/9$ for the second and third tasks. This algebraic asymmetry is present before any algorithmic mechanism is considered.

For A-first runs, mean initial FDE is 21.52 m, compared with 1.04–2.77 m for other starting tasks. The subsequent-error and reference terms are 1.46 and 13.69 m, respectively, yielding BWT $= -12.23$ m despite a final mean error of 1.28 m. Table VI reports these quantities by starting task.

Across algorithms, the correlation of D with the FDE reference term B is between -0.69 and -0.57, whereas its correlation with the subsequent-error term P is only $+0.02$ to $+0.19$. The stronger association with FDE-BWT is consistent

TABLE II
CURRENT-TASK ERRORS BY TASK ORDER (MEAN ± SAMPLE SD). FDE IN METRES; MR IN PERCENT.

Order	Vanilla		DER		SyReM		GSM	
	FDE_{CT}	MR_{CT}	FDE_{CT}	MR_{CT}	FDE_{CT}	MR_{CT}	FDE_{CT}	MR_{CT}
1	$8.25_{\pm 0.46}$	$32.83_{\pm 2.59}$	$6.17_{\pm 2.13}$	$31.75_{\pm 1.97}$	$7.17_{\pm 3.71}$	$31.74_{\pm 1.65}$	$6.08_{\pm 1.36}$	$31.84_{\pm 0.56}$
2	$4.98_{\pm 2.49}$	$30.83_{\pm 4.43}$	$6.67_{\pm 2.39}$	$32.20_{\pm 1.14}$	$6.55_{\pm 2.48}$	$28.76_{\pm 2.27}$	$8.00_{\pm 3.49}$	$34.38_{\pm 0.52}$
3	$7.33_{\pm 3.01}$	$33.28_{\pm 4.02}$	$6.69_{\pm 2.57}$	$29.44_{\pm 1.99}$	$3.31_{\pm 0.31}$	$25.33_{\pm 0.67}$	$6.23_{\pm 1.09}$	$31.86_{\pm 2.69}$
4	$9.26_{\pm 1.95}$	$29.96_{\pm 1.81}$	$3.48_{\pm 1.02}$	$28.85_{\pm 3.15}$	$4.71_{\pm 1.20}$	$27.92_{\pm 4.05}$	$5.00_{\pm 1.84}$	$30.21_{\pm 1.17}$
5	$9.20_{\pm 1.98}$	$31.51_{\pm 2.50}$	$3.38_{\pm 0.70}$	$28.18_{\pm 4.41}$	$3.90_{\pm 1.97}$	$26.08_{\pm 6.67}$	$9.17_{\pm 2.11}$	$35.83_{\pm 0.44}$
6	$5.92_{\pm 2.54}$	$26.34_{\pm 4.40}$	$5.10_{\pm 3.73}$	$25.08_{\pm 1.65}$	$6.53_{\pm 2.44}$	$29.38_{\pm 0.54}$	$6.97_{\pm 0.34}$	$32.37_{\pm 2.82}$
7	$2.40_{\pm 1.13}$	$21.10_{\pm 8.23}$	$2.97_{\pm 1.77}$	$21.92_{\pm 13.62}$	$2.16_{\pm 0.20}$	$21.02_{\pm 3.63}$	$2.66_{\pm 1.38}$	$21.34_{\pm 10.22}$
8	$2.13_{\pm 0.63}$	$20.75_{\pm 6.81}$	$2.09_{\pm 0.65}$	$19.11_{\pm 8.98}$	$1.61_{\pm 0.48}$	$15.98_{\pm 5.36}$	$2.70_{\pm 1.27}$	$23.88_{\pm 7.66}$
9	$1.45_{\pm 0.37}$	$12.31_{\pm 4.21}$	$2.14_{\pm 1.02}$	$22.57_{\pm 13.92}$	$1.21_{\pm 0.55}$	$8.46_{\pm 4.87}$	$1.52_{\pm 0.32}$	$13.24_{\pm 4.24}$
10	$1.21_{\pm 0.32}$	$9.16_{\pm 1.51}$	$1.12_{\pm 0.30}$	$7.34_{\pm 2.98}$	$1.25_{\pm 0.22}$	$10.65_{\pm 2.05}$	$1.72_{\pm 0.57}$	$16.51_{\pm 9.91}$
11	$1.62_{\pm 0.19}$	$15.08_{\pm 3.63}$	$1.75_{\pm 0.41}$	$13.61_{\pm 1.82}$	$1.48_{\pm 0.28}$	$15.67_{\pm 7.44}$	$1.80_{\pm 0.50}$	$16.02_{\pm 7.99}$
12	$1.53_{\pm 0.53}$	$11.71_{\pm 3.11}$	$1.33_{\pm 0.54}$	$12.87_{\pm 7.73}$	$1.63_{\pm 0.48}$	$12.85_{\pm 3.69}$	$1.84_{\pm 0.19}$	$17.31_{\pm 3.29}$
13	$1.27_{\pm 0.28}$	$14.74_{\pm 5.90}$	$1.13_{\pm 0.25}$	$10.49_{\pm 4.61}$	$1.28_{\pm 0.48}$	$14.34_{\pm 9.44}$	$1.01_{\pm 0.20}$	$7.81_{\pm 3.57}$
14	$1.66_{\pm 0.42}$	$22.08_{\pm 6.90}$	$1.07_{\pm 0.35}$	$9.75_{\pm 7.29}$	$1.15_{\pm 0.23}$	$11.32_{\pm 5.51}$	$0.97_{\pm 0.15}$	$7.47_{\pm 3.60}$
15	$1.07_{\pm 0.10}$	$9.64_{\pm 1.13}$	$1.07_{\pm 0.30}$	$6.78_{\pm 2.11}$	$0.94_{\pm 0.08}$	$7.34_{\pm 2.35}$	$1.58_{\pm 0.36}$	$17.08_{\pm 8.05}$
16	$1.02_{\pm 0.11}$	$8.37_{\pm 1.34}$	$1.12_{\pm 0.37}$	$11.67_{\pm 8.13}$	$1.00_{\pm 0.15}$	$8.76_{\pm 3.74}$	$1.05_{\pm 0.26}$	$7.85_{\pm 3.42}$
17	$1.32_{\pm 0.79}$	$12.34_{\pm 10.76}$	$0.91_{\pm 0.13}$	$6.68_{\pm 3.23}$	$1.22_{\pm 0.63}$	$9.62_{\pm 8.52}$	$1.16_{\pm 0.20}$	$11.07_{\pm 2.88}$
18	$1.05_{\pm 0.34}$	$7.88_{\pm 3.92}$	$0.95_{\pm 0.03}$	$7.58_{\pm 0.53}$	$0.89_{\pm 0.09}$	$5.91_{\pm 2.21}$	$0.99_{\pm 0.18}$	$8.20_{\pm 4.24}$
19	$1.76_{\pm 0.93}$	$18.29_{\pm 13.10}$	$1.08_{\pm 0.20}$	$10.46_{\pm 4.87}$	$1.84_{\pm 1.25}$	$17.54_{\pm 11.58}$	$2.04_{\pm 0.82}$	$21.88_{\pm 10.08}$
20	$1.75_{\pm 0.12}$	$23.72_{\pm 5.06}$	$1.14_{\pm 0.17}$	$13.03_{\pm 4.88}$	$1.37_{\pm 0.59}$	$12.03_{\pm 4.42}$	$2.92_{\pm 2.14}$	$23.69_{\pm 7.29}$
21	$1.27_{\pm 0.50}$	$9.68_{\pm 3.12}$	$2.39_{\pm 1.76}$	$21.68_{\pm 11.80}$	$1.36_{\pm 0.53}$	$14.11_{\pm 9.07}$	$1.49_{\pm 0.56}$	$19.62_{\pm 11.26}$
22	$1.14_{\pm 0.18}$	$13.15_{\pm 3.24}$	$1.10_{\pm 0.16}$	$10.39_{\pm 3.88}$	$1.18_{\pm 0.23}$	$11.06_{\pm 1.64}$	$1.36_{\pm 0.45}$	$15.95_{\pm 7.94}$
23	$1.04_{\pm 0.06}$	$10.12_{\pm 1.57}$	$2.24_{\pm 0.90}$	$24.82_{\pm 7.94}$	$1.01_{\pm 0.13}$	$7.89_{\pm 2.05}$	$1.02_{\pm 0.11}$	$9.98_{\pm 2.72}$
24	$1.01_{\pm 0.08}$	$9.44_{\pm 2.96}$	$1.03_{\pm 0.13}$	$9.01_{\pm 2.85}$	$1.06_{\pm 0.10}$	$11.17_{\pm 2.48}$	$1.32_{\pm 0.52}$	$14.56_{\pm 7.85}$

TABLE III
BACKWARD TRANSFER BY TASK ORDER (MEAN ± SAMPLE SD). FDE IN METRES; MR IN PERCENTAGE POINTS.

Order	Vanilla		DER		SyReM		GSM	
	FDE_{BWT}	MR_{BWT}	FDE_{BWT}	MR_{BWT}	FDE_{BWT}	MR_{BWT}	FDE_{BWT}	MR_{BWT}
1	$-16.39_{\pm 1.57}$	$-46.19_{\pm 3.15}$	$-10.99_{\pm 5.05}$	$-41.41_{\pm 0.95}$	$-14.77_{\pm 8.97}$	$-52.95_{\pm 2.18}$	$-12.00_{\pm 3.63}$	$-54.18_{\pm 2.86}$
2	$-8.41_{\pm 5.26}$	$-42.47_{\pm 2.39}$	$-12.28_{\pm 5.50}$	$-44.11_{\pm 2.93}$	$-13.55_{\pm 5.89}$	$-54.58_{\pm 3.96}$	$-15.61_{\pm 9.07}$	$-48.23_{\pm 11.43}$
3	$-13.97_{\pm 7.56}$	$-47.29_{\pm 3.65}$	$-13.49_{\pm 6.02}$	$-46.46_{\pm 2.77}$	$-5.96_{\pm 0.70}$	$-50.20_{\pm 1.11}$	$-12.35_{\pm 2.34}$	$-49.80_{\pm 3.44}$
4	$-19.84_{\pm 4.77}$	$-42.47_{\pm 11.19}$	$-5.36_{\pm 2.77}$	$-37.45_{\pm 12.64}$	$-9.18_{\pm 3.12}$	$-53.14_{\pm 5.23}$	$-8.97_{\pm 5.24}$	$-48.22_{\pm 7.97}$
5	$-19.42_{\pm 4.94}$	$-45.86_{\pm 2.94}$	$-5.44_{\pm 1.35}$	$-40.02_{\pm 8.01}$	$-7.15_{\pm 4.53}$	$-45.12_{\pm 8.76}$	$-18.75_{\pm 5.52}$	$-50.08_{\pm 2.73}$
6	$-11.50_{\pm 6.68}$	$-33.69_{\pm 21.46}$	$-10.01_{\pm 9.22}$	$-41.18_{\pm 7.96}$	$-13.73_{\pm 5.86}$	$-56.97_{\pm 0.56}$	$-14.32_{\pm 0.59}$	$-49.42_{\pm 2.57}$
7	$0.38_{\pm 1.30}$	$-4.77_{\pm 2.03}$	$0.38_{\pm 0.41}$	$2.80_{\pm 7.91}$	$-0.35_{\pm 0.64}$	$-12.22_{\pm 2.50}$	$-0.68_{\pm 1.66}$	$-8.45_{\pm 10.69}$
8	$-0.95_{\pm 0.77}$	$-8.34_{\pm 10.34}$	$-0.13_{\pm 0.46}$	$-3.38_{\pm 7.23}$	$-0.18_{\pm 1.07}$	$-7.88_{\pm 12.83}$	$-0.36_{\pm 1.37}$	$-5.76_{\pm 7.80}$
9	$0.17_{\pm 0.84}$	$4.50_{\pm 2.97}$	$0.57_{\pm 0.80}$	$7.14_{\pm 4.99}$	$-0.51_{\pm 1.00}$	$-4.43_{\pm 9.09}$	$-0.63_{\pm 0.44}$	$-3.01_{\pm 2.01}$
10	$0.97_{\pm 1.18}$	$12.43_{\pm 9.35}$	$1.08_{\pm 0.29}$	$10.74_{\pm 3.78}$	$-0.47_{\pm 0.65}$	$-3.43_{\pm 7.03}$	$-0.72_{\pm 0.92}$	$1.58_{\pm 3.47}$
11	$0.44_{\pm 0.85}$	$4.86_{\pm 1.12}$	$-0.01_{\pm 0.80}$	$5.39_{\pm 4.17}$	$-0.38_{\pm 0.66}$	$-5.72_{\pm 3.49}$	$-0.85_{\pm 0.32}$	$-2.73_{\pm 2.70}$
12	$0.43_{\pm 1.25}$	$7.10_{\pm 11.32}$	$0.87_{\pm 1.00}$	$11.62_{\pm 2.77}$	$-1.02_{\pm 1.18}$	$-0.11_{\pm 3.79}$	$-0.81_{\pm 0.70}$	$1.27_{\pm 4.80}$
13	$0.66_{\pm 0.14}$	$6.36_{\pm 4.90}$	$0.68_{\pm 0.13}$	$9.91_{\pm 3.58}$	$-0.10_{\pm 0.38}$	$-2.38_{\pm 7.88}$	$0.22_{\pm 0.05}$	$5.57_{\pm 1.98}$
14	$0.38_{\pm 0.15}$	$3.60_{\pm 5.30}$	$0.45_{\pm 0.14}$	$8.91_{\pm 3.94}$	$0.05_{\pm 0.12}$	$1.92_{\pm 3.45}$	$0.42_{\pm 0.30}$	$8.87_{\pm 6.16}$
15	$0.97_{\pm 0.31}$	$11.47_{\pm 3.74}$	$0.80_{\pm 0.19}$	$14.11_{\pm 1.94}$	$0.11_{\pm 0.17}$	$0.22_{\pm 4.51}$	$0.46_{\pm 0.63}$	$10.30_{\pm 4.48}$
16	$1.12_{\pm 0.19}$	$13.35_{\pm 3.41}$	$1.29_{\pm 0.34}$	$15.87_{\pm 2.03}$	$-0.07_{\pm 0.25}$	$-2.14_{\pm 6.78}$	$0.26_{\pm 0.37}$	$7.52_{\pm 6.05}$
17	$0.64_{\pm 0.24}$	$14.77_{\pm 3.67}$	$0.66_{\pm 0.18}$	$14.69_{\pm 2.58}$	$0.21_{\pm 0.25}$	$3.29_{\pm 3.01}$	$0.58_{\pm 0.13}$	$13.43_{\pm 1.53}$
18	$0.82_{\pm 0.12}$	$16.91_{\pm 1.55}$	$0.73_{\pm 0.48}$	$13.16_{\pm 8.55}$	$0.21_{\pm 0.15}$	$4.24_{\pm 3.76}$	$0.85_{\pm 0.51}$	$15.54_{\pm 6.66}$
19	$0.40_{\pm 0.45}$	$10.65_{\pm 7.35}$	$0.71_{\pm 0.16}$	$14.06_{\pm 2.05}$	$-0.49_{\pm 1.34}$	$-3.59_{\pm 12.00}$	$1.38_{\pm 1.66}$	$15.06_{\pm 9.24}$
20	$0.45_{\pm 0.07}$	$5.37_{\pm 3.07}$	$0.64_{\pm 0.18}$	$11.91_{\pm 4.44}$	$0.29_{\pm 0.38}$	$2.62_{\pm 1.80}$	$-0.75_{\pm 1.53}$	$0.84_{\pm 2.56}$
21	$0.86_{\pm 0.49}$	$17.83_{\pm 2.77}$	$1.04_{\pm 0.34}$	$15.53_{\pm 2.89}$	$0.21_{\pm 0.06}$	$3.21_{\pm 1.64}$	$0.90_{\pm 0.16}$	$8.44_{\pm 5.08}$
22	$1.03_{\pm 0.17}$	$17.66_{\pm 3.60}$	$0.95_{\pm 0.14}$	$16.63_{\pm 3.17}$	$0.01_{\pm 0.19}$	$2.55_{\pm 1.02}$	$0.84_{\pm 0.58}$	$13.79_{\pm 5.05}$
23	$0.73_{\pm 0.11}$	$15.61_{\pm 1.74}$	$0.73_{\pm 0.37}$	$12.99_{\pm 5.99}$	$0.21_{\pm 0.22}$	$6.13_{\pm 6.09}$	$0.54_{\pm 0.25}$	$10.47_{\pm 4.67}$
24	$0.85_{\pm 0.32}$	$15.16_{\pm 6.63}$	$0.91_{\pm 0.27}$	$18.19_{\pm 3.59}$	$0.03_{\pm 0.12}$	$-0.22_{\pm 4.14}$	$0.53_{\pm 0.41}$	$11.18_{\pm 5.78}$

with a substantial contribution from the order-dependent reference. This finding weakens an interpretation of the marginal correlation as direct evidence that larger D harms retention.

The final-error comparison in Table VII yields a different algorithm ranking. SyReM has the lowest uniform-order final FDE (1.08 m) and MR (9.23%), whereas GSM has the smallest uniform-order FDE-BWT. Thus, improvement relative to an algorithm-specific initial checkpoint and absolute final prediction quality induce distinct comparisons.

CT and BWT contain common initial-error terms with opposite signs; their opposite associations with D do not by themselves establish an optimisation trade-off. Final error provides an absolute comparison. Isolating a mechanism would additionally require control of initial fit, training exposure, and task size.

D. Different Paths from the Same Starting Task

Fixing the first task still leaves six possible orders of the remaining three. We select the largest- and smallest-L path within each group using the symmetric matrix of Wasserstein

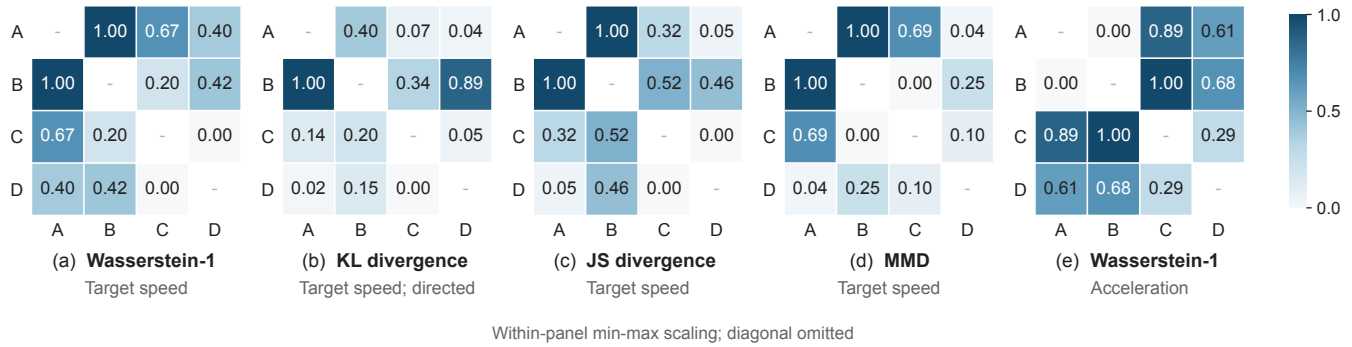


Fig. 5. Candidate task-distance matrices, scaled independently to $[0, 1]$ for display. A–D denote the scenario groups. KL is directed; the other matrices are symmetric. Speed Wasserstein defines the primary descriptor.

TABLE IV
SPEARMAN CORRELATIONS FOR FIVE CANDIDATE TRANSITION DESCRIPTORS. BOLD: $|\rho| \geq 0.7$.

Algorithm	Metric	W1 speed	KL speed	JS speed	MMD speed	W1 accel.
Vanilla	FDE_{CT}	**-0.75**	-0.23	-0.20	**-0.75**	+0.69
	MR_{CT}	**-0.75**	-0.09	-0.12	**-0.75**	+0.69
	FDE_{BWT}	+0.66	+0.13	+0.10	+0.66	-0.55
	MR_{BWT}	**+0.81**	+0.26	+0.32	**+0.81**	-0.46
DER	FDE_{CT}	-0.57	-0.29	-0.36	-0.57	+0.46
	MR_{CT}	-0.54	-0.22	-0.27	-0.54	+0.48
	FDE_{BWT}	+0.67	+0.10	+0.12	+0.67	-0.52
	MR_{BWT}	**+0.74**	+0.28	+0.32	**+0.74**	-0.43
SyReM	FDE_{CT}	-0.67	-0.29	-0.26	-0.67	**+0.74**
	MR_{CT}	**-0.72**	-0.22	-0.23	**-0.72**	**+0.76**
	FDE_{BWT}	+0.55	+0.31	+0.36	+0.55	-0.35
	MR_{BWT}	**+0.72**	+0.35	+0.43	**+0.72**	-0.42
GSM	FDE_{CT}	-0.57	-0.30	-0.29	-0.57	+0.66
	MR_{CT}	-0.53	-0.23	-0.23	-0.53	+0.69
	FDE_{BWT}	+0.55	+0.32	+0.30	+0.55	-0.37
	MR_{BWT}	+0.69	+0.40	+0.44	+0.69	-0.43
Mean $\|\rho\|$, CT		0.64	0.23	0.24	0.64	0.65
Mean $\|\rho\|$, BWT		0.67	0.27	0.30	0.67	0.44
Mean $\|\rho\|$, all		0.66	0.25	0.27	0.66	0.54

TABLE V
MEAN DESCRIPTOR CORRELATIONS ACROSS ALGORITHMS, WITH AND WITHOUT ADJUSTMENT FOR THE STARTING TASK.

Metric	All orders	Excluding A-first	Adjusted
FDE_{CT}	-0.64	-0.47	-0.48
MR_{CT}	-0.63	-0.43	-0.49
FDE_{BWT}	+0.61	+0.40	+0.29
MR_{BWT}	+0.74	+0.64	+0.62

TABLE VI
FDE DECOMPOSITION BY STARTING TASK (METRES), AVERAGED OVER 72 TRAINING RUNS PER ROW.

Start	$R_{1,1}$	P	B	BWT	Final mean
A	21.52	1.46	13.69	-12.23	1.28
B	2.77	2.34	2.46	-0.12	1.52
C	1.04	1.64	1.13	0.52	1.50
D	1.32	2.04	1.50	0.54	1.48

TABLE VII
FINAL ERRORS: MEAN [MINIMUM, MAXIMUM] OVER ORDER-LEVEL REPLICATE MEANS. FDE IN METRES; MR IN PERCENT.

Algorithm	FDE: mean [min, max]	MR: mean [min, max]
Vanilla	1.63 [1.18, 2.14]	17.22 [10.84, 25.67]
DER	1.63 [1.17, 2.34]	17.80 [11.47, 29.09]
SyReM	1.08 [0.87, 1.99]	9.23 [4.38, 24.69]
GSM	1.43 [1.03, 2.44]	15.86 [8.05, 36.25]

distances on speed. Fig. 7 shows the selected sequences and their measured performance. Fixing the start provides a common initial task while allowing the later transitions to differ.

For all four starting tasks, the maximum-distance path has a smaller D than the corresponding minimum-distance path. The selected path pairs have different measured performance, but the direction of the difference is not universal across metrics. For example, the two B-start paths have algorithm-averaged FDE-CT values of 2.55 and 1.33 m. This comparison

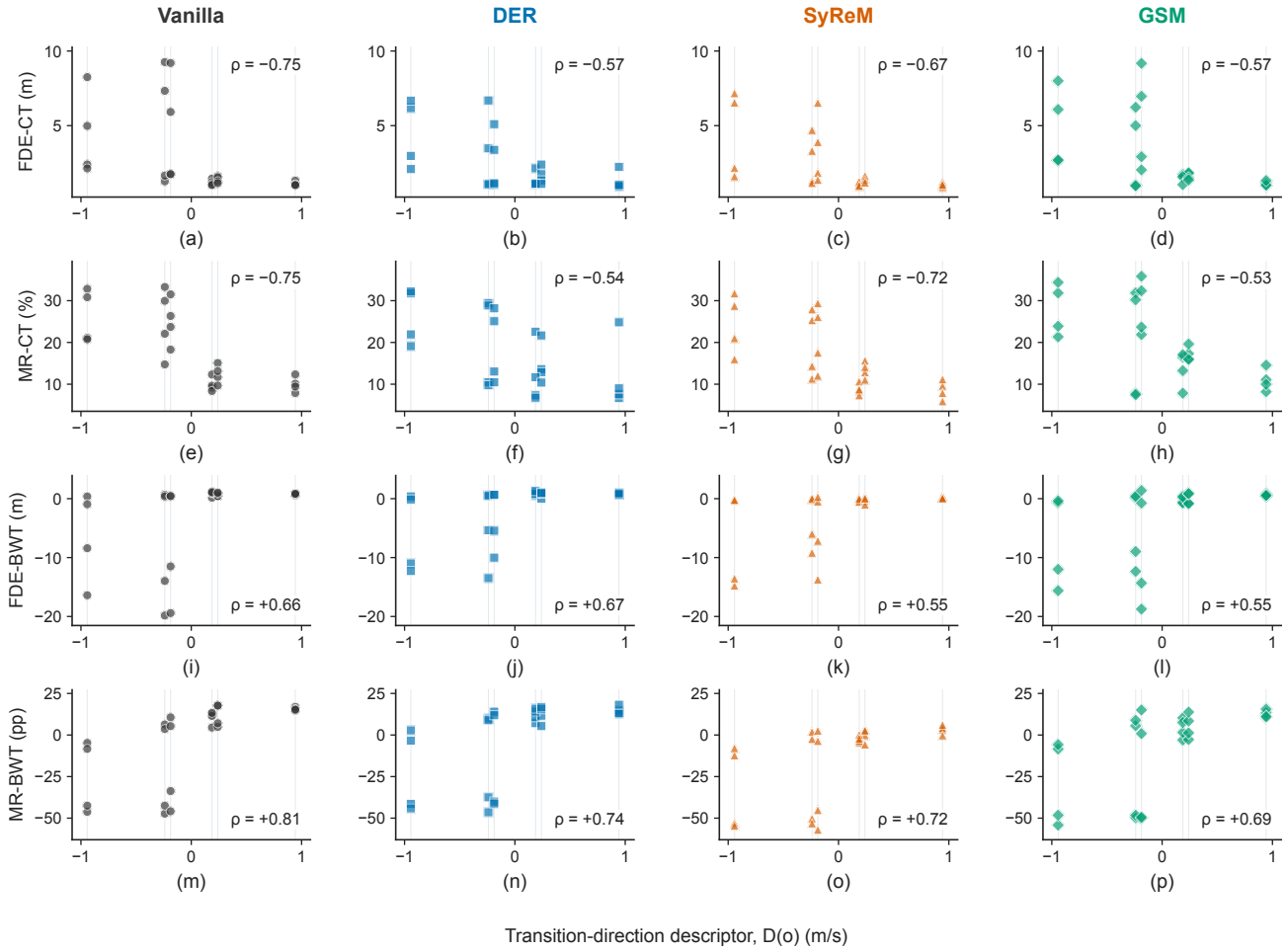


Fig. 6. Task-transition descriptor versus performance for each algorithm. Points are replicate means for individual orders; ρ is Spearman correlation. Grey guides mark the distinct descriptor values.

demonstrates sensitivity beyond starting identity while leaving the source of that sensitivity unresolved.

The largest- and smallest-distance paths are geometric extremes under L, not upper and lower bounds on CL performance. Neither their scores nor their average is guaranteed to bracket the uniform or weighted evaluation. Their role is to provide interpretable, explicitly selected sequence comparisons alongside the complete order-resolved results.

E. Effect of the Evaluation Policy

Table VIII compares the uniform and D-weighted scores. Weighting increases the reported CT error for every algorithm: SyReM's FDE-CT changes from 2.33 to 3.01 m, while Vanilla's changes from 2.94 to 3.76 m. BWT also shifts upward, for example from -13.79 to -6.04 percentage points for SyReM's MR-BWT. These changes follow the empirical association used to choose the weighting directions and should not be counted as independent confirmation of that association.

The algorithm with the smallest mean in each metric is unchanged under the two policies: SyReM for FDE-CT, MR-CT, and MR-BWT, and GSM for FDE-BWT. The contribution of weighting on this benchmark is therefore to show how the numerical comparison changes under a stated order emphasis, not to demonstrate improved selection of the best algorithm. The effective number of weighted orders is 17.54 for either policy, compared with 24 under uniform weights. This quantity describes weight concentration, not statistical independence.

The weighted score estimates performance under an alternative order policy that emphasises a specified region of the descriptor; the uniform mean retains the equal-order reference.

TABLE VIII
UNIFORM AND CONTRAST-WEIGHTED SCORES. BOLD MARKS THE SMALLEST VALUE IN EACH ROW.

Metric	Weight	Vanilla	DER	SyReM	GSM
$\mathrm{FDE_{CT}}$	Uniform	2.94	2.42	**2.33**	2.94
	D-weighted	3.76	3.06	**3.01**	3.76
$\mathrm{MR_{CT}}$	Uniform	18.10	17.30	**16.04**	19.58
	D-weighted	21.87	20.12	**19.34**	22.98
$\mathrm{FDE_{BWT}}$	Uniform	-3.30	-1.88	-2.77	**-3.33**
	D-weighted	-1.46	-0.43	-1.23	**-1.55**
$\mathrm{MR_{BWT}}$	Uniform	-3.89	-2.10	**-13.79**	-8.17
	D-weighted	5.17	6.01	**-6.04**	0.74



Fig. 7. Paths with maximum and minimum total transition distance for each starting task. Sequence strips identify the paths; paired points compare algorithm-averaged scores, standardised separately for each metric across task orders.

Choosing either policy after seeing which algorithm it favours would introduce another selection effect. For comparisons on new task sets, the descriptor, direction, and weighting rule should therefore be fixed before examining algorithm outcomes.

VII. Discussion

A. Interpreting Effects of Task Order

Algorithm rankings vary with learning history, making the evaluated order set part of the benchmark definition. The starting task has a strong influence: its initial error enters CT and recurs in the reference term of stage-averaged BWT. The large A-first reference helps explain the opposite marginal associations of CT and BWT with the transition descriptor. The distinct leaders under final error and FDE-BWT further demonstrate that acquisition, subsequent change, and final predictive quality require separate assessment.

B. Implications for Algorithm Comparison

A stated order distribution defines the interpretation of an aggregate score. Uniform weighting provides an equal-order reference; alternative policies can represent anticipated deployment conditions or emphasise particular transition patterns. In this experiment, descriptor weighting changes the aggregate scores while preserving the leading algorithm for each metric. Stability of this aggregate ranking therefore coexists with ranking reversals on individual sequences.

A practical report should combine average scores, per-order differences, and training variation. Ranks describe the frequency of an algorithmic advantage, whereas error gaps describe its magnitude. Final error contextualises transfer metrics, and comparisons under shared order policies quantify the dependence of aggregate results on the evaluation design.

C. Towards a Task-general Evaluation Protocol

A principal direction for future work is a general and fair order-aware evaluation protocol spanning different task definitions, application domains, and continual learning paradigms. Such a protocol should specify task partitions, learning and memory budgets, and a common order distribution before algorithm outcomes are observed. The evaluation policy should be independent of the method being assessed, while the coverage of task positions and transitions should be reported explicitly.

For larger task sets, sampling from a declared order distribution can replace complete enumeration. Balanced coverage of starting tasks and transition types offers a candidate design; repeated order samples would quantify coverage uncertainty separately from training variability. Validation across independently constructed task sets, architectures, and prediction objectives would determine whether an evaluation policy transfers beyond motion forecasting. Developing task representations that support these comparisons without requiring application-specific features, such as vehicle speed, is central to this extension. The present study provides a concrete implementation on a fixed forecasting benchmark; a task-general protocol requires evidence across these broader settings.

VIII. Conclusion

This study formulates continual learning evaluation under an explicit distribution over task orders. Shared-sequence comparisons reveal ranking reversals and substantial variation across learning histories. A descriptor of task distributions provides a common weighting policy that complements uniform evaluation, while adjustment for the starting task and decomposition of backward transfer distinguish order associations from initial-reference effects. The findings support

reporting order-resolved outcomes and absolute final errors alongside aggregate scores. Extending this approach to a task-general, fair benchmarking protocol requires shared sampling principles and validation across independently constructed task sets.

APPENDIX A
ANALYSIS DETAILS AND REPRODUCIBILITY

A. Feature and Distance Reconstruction

Target-agent velocity components are extracted at nine historical steps from each trajectory. Speed is $\sqrt{v_x^2 + v_y^2}$ and acceleration is its first difference divided by 0.1 s, retaining values in $[-5, 5]$ m/s^2. Both channels match the cached arrays element by element across all twelve subsets. The release includes feature counts, source hashes, and these verification results.

Equations (12)–(18) specify the distance calculations. The pooled mean and population standard deviation in Eq. (16) use all retained speed observations. MMD samples are drawn without replacement using NumPy PCG64 with seed 0, proceeding through tasks A–D in order. Exact bin edges, standardisation constants, and sampled indices accompany the data.

Historical matrices are retained for sensitivity analysis. The original speed-Wasserstein matrix is symmetrised as $(M + M^\top)/2$; it induces the same descriptor ranking as the reconstruction. Display scaling is excluded from all calculations.

B. Descriptive Variation and Conditional Association

For a fixed metric, let Z_{oas} be a run outcome, with $K = |\mathcal{O}|$, $J = |\mathcal{A}|$, and S replicates per combination. Dots denote averaging over the corresponding indices and $\bar{Z}$ is the grand mean. The balanced decomposition is

$$SS_O = JS \sum_o (\bar{Z}_{o\cdot\cdot} - \bar{Z})^2, \tag{30}$$

$$SS_A = KS \sum_a (\bar{Z}_{\cdot a\cdot} - \bar{Z})^2, \tag{31}$$

$$SS_{OA} = S \sum_{o,a} (\bar{Z}_{oa\cdot} - \bar{Z}_{o\cdot\cdot} - \bar{Z}_{\cdot a\cdot} + \bar{Z})^2, \tag{32}$$

$$SS_W = \sum_{o,a,s} (Z_{oas} - \bar{Z}_{oa\cdot})^2. \tag{33}$$

These terms sum to $SS_T = \sum_{o,a,s} (Z_{oas} - \bar{Z})^2$. The within-combination term SS_W captures training variation and is invariant to replicate relabelling within each combination. The reported fractions are descriptive rather than causal variance components.

For each algorithm–metric series, conditional rank association is computed by assigning average ranks to D and to the order mean, subtracting the respective mean rank within each starting-task group, and correlating the two residual vectors. All evaluated orders contribute to this calculation. It is distinct from omitting A-first orders or correlating raw-value residuals. All candidate associations are reported without post-selection significance claims.

C. Weighting and Edge Cases

The complete order set includes the reversal of every order. For a nonconstant symmetric-distance descriptor, paired values $D(o)$ and $-D(o)$ imply $D_{\min} = -D_{\max}$. Consequently, both CT and BWT weights sum to $|\mathcal{O}|/2$. The effective number of weighted order means, $(\sum_o w_o)^2 / \sum_o w_o^2$, describes weight concentration only; it is not an effective number of independent task sets. The constant-descriptor case uses uniform weights. All reported weighted means are convex combinations and remain between the smallest and largest measured order means.

APPENDIX B
TASK ORDER IDENTIFIERS

Table IX groups sequences by their first task: A, Urban Merge; B, Highway Merge; C, Intersection; and D, Roundabout.

TABLE IX
TASK ORDER IDENTIFIERS GROUPED BY STARTING TASK.

Start A		Start B		Start C		Start D	
ID	Sequence	ID	Sequence	ID	Sequence	ID	Sequence
1	ABCD	7	BACD	13	CABD	19	DABC
2	ABDC	8	BADC	14	CADB	20	DACB
3	ACBD	9	BCAD	15	CBAD	21	DBAC
4	ACDB	10	BCDA	16	CBDA	22	DBCA
5	ADBC	11	BDAC	17	CDAB	23	DCAB
6	ADCB	12	BDCA	18	CDBA	24	DCBA

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